Exp 11
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# Routing at Network layer

Aim :-

To simulate static routing protocol configuration and RIP using Cisco Packet Tracer.

Algorithm :-

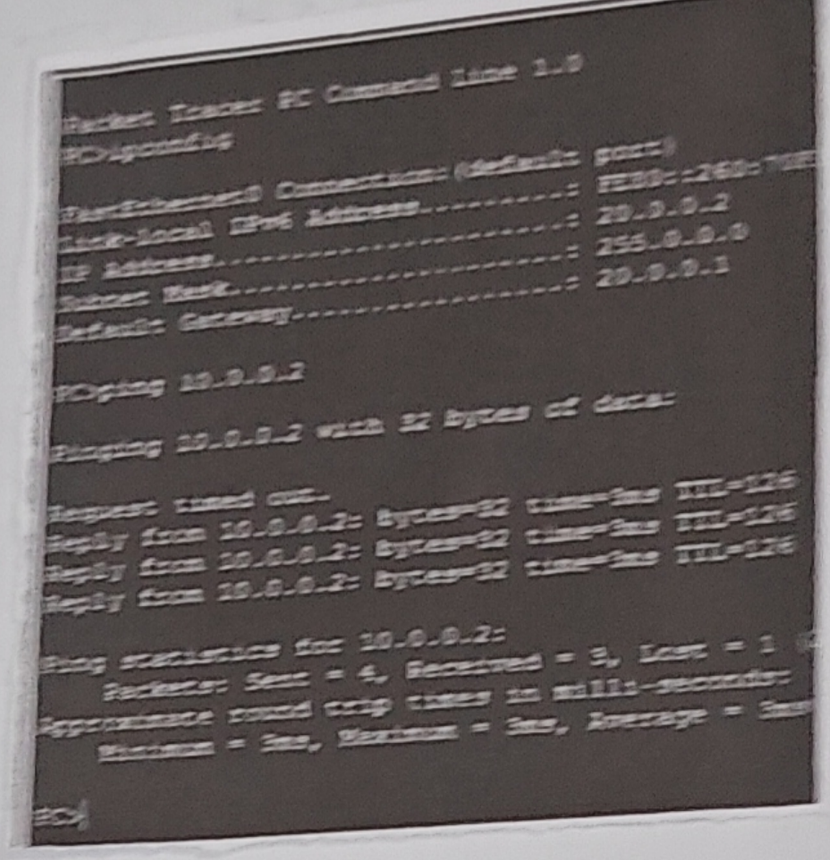
⇒ Part A - Static Routing

1. Open cisco packet trace & create given network topology with routers
2. Routers learn networks available on own interfaces
3. Open router CLI & enter mode.
4. First route is configured as main route & 2nd is Backup route
5. Use tracert command to track the path taken by packets
6. To test Backup route, remove link b/w router 0 and router 1
7. A static route can be deleted using no ip route

⇒ Part B - RIP using Cisco Packet Tracer

configure IP address of PCs & network interfaces acc. to given network.

configure serial interfaces of router with given IP address.


Packet Tracer PC Command Line 1.0
PC>ipconfig
FastEthernet Connection (default port)
Link-local IPv6 Address.....:
IP Address.....: 20.0.0.2
Subnet Mask.....: 255.0.0.0
Default Gateway.....: 20.0.0.1
PC>ping 10.0.0.2
Pinging 10.0.0.2 with 32 bytes of data:
Request timed out.
Reply from 10.0.0.2: bytes=32 time=1ms TTL=126
Reply from 10.0.0.2: bytes=32 time=1ms TTL=126
Reply from 10.0.0.2: bytes=32 time=1ms TTL=126
Ping statistics for 10.0.0.2:
Packets: Sent = 4, Received = 3, Lost = 1
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 1ms, Average = 1
PC>

2. configure & enable RIP routing protocol from global configuration mode.

4. specify directly connected networks to be advertised.

5. use ping command to test connectivity b/w PC0 & PC1

6. use tracert command

7. If selected route fails, RIP reroutes traffic through another available rou

Result:-

static routing and RIP were simulated using Cisco Packet Tracer & packet transmission and alternate routing verified.